Maximum Score Routing For Mixture-of-Experts

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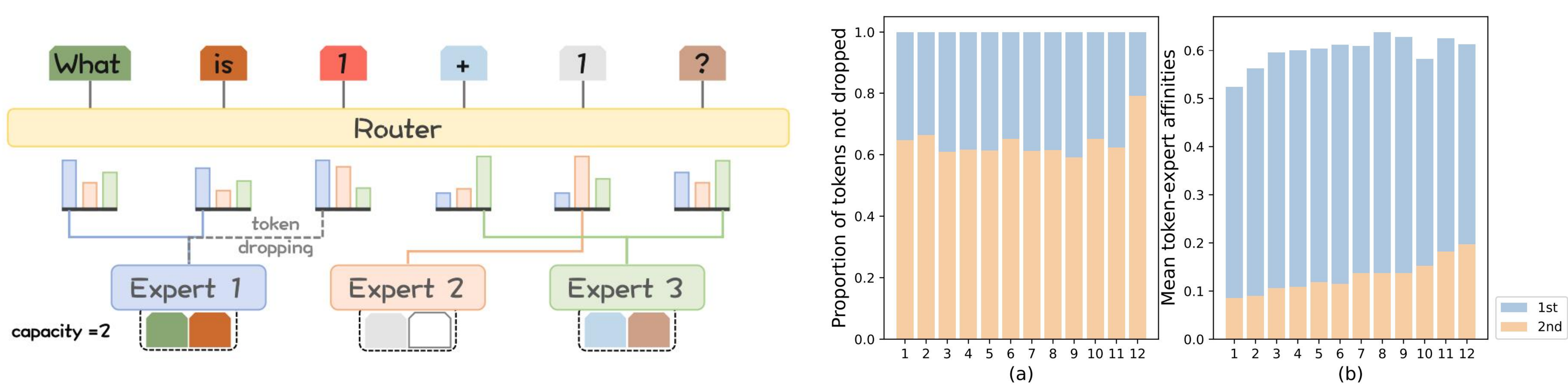
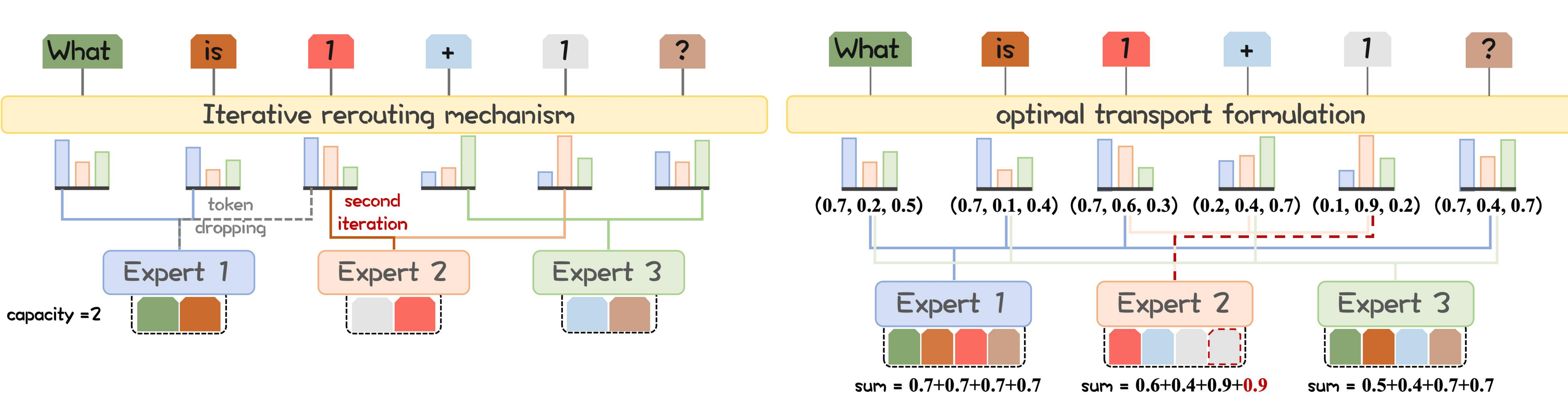
❖ Motivation

- In a Top-2 MoE routing of GShard, the probability of **Token Dropping** for the second tokens is 35%.
- Since **Softmax** is used to calculate the token-expert affinity coefficients, this makes the probability distribution close to the **One-hot** distribution.
- The probability of top1 is significantly higher than that of top2.

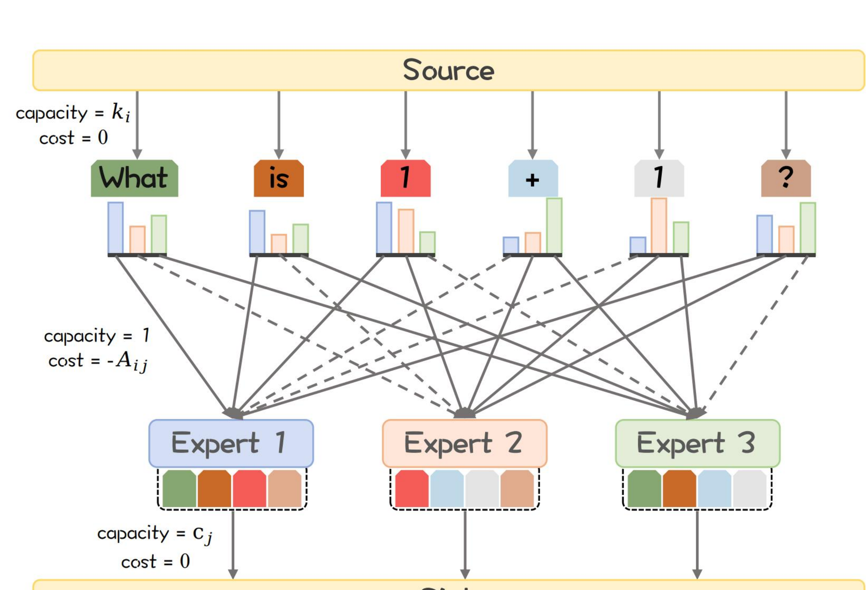


- Switch Transformer attempted to recycle the dropping tokens using the Iterative method, but found no benefit to model quality.
- S-BASE models the MoE routing as an optimal Transport (OT) problem and uses the Sinkhorn algorithm for routing. No benefit, either.



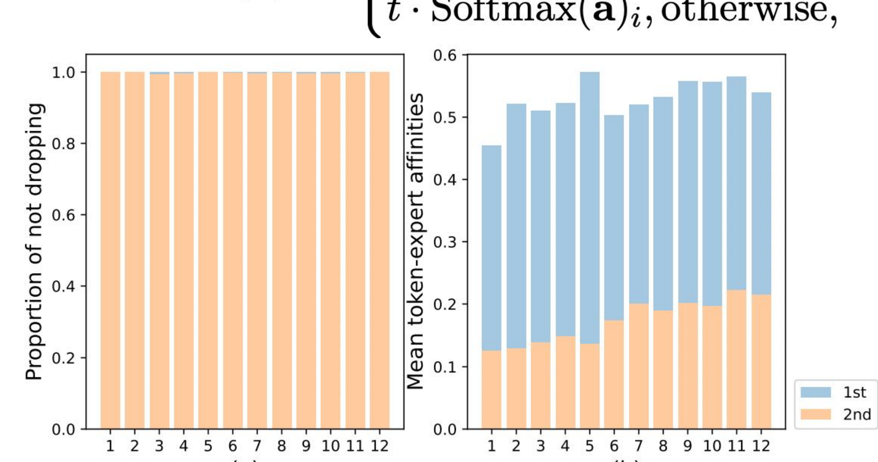
❖ Methodology

- This work introduces **Maximum Score Routing** (**MaxScore**), a novel MoE routing paradigm that formulates token-expert routing as a **Minimum-Cost Maximum-Flow** problem, integrated with a **SoftTopk** operator.
- By introducing the **SoftTopk** operator, the distribution of the expert-token affinity coefficient is closer to **K-shot** rather than One-shot.

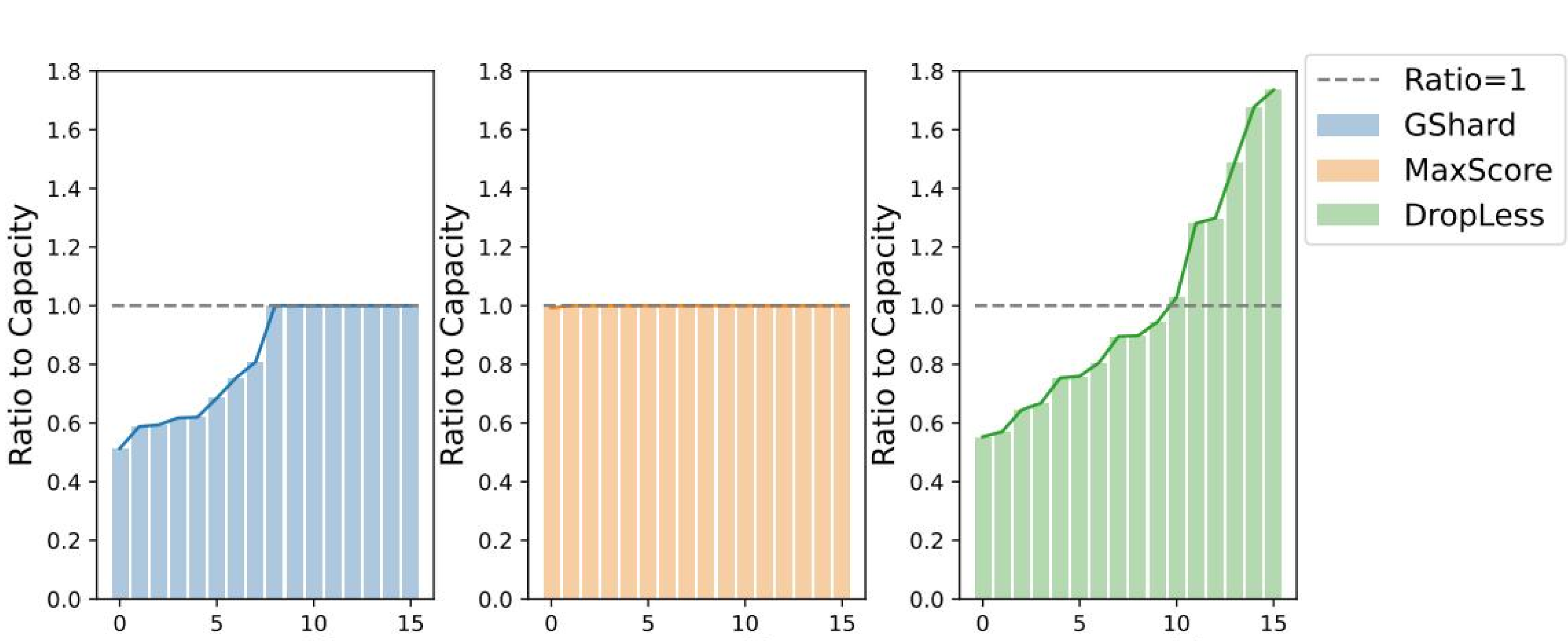


$$\mathrm{SoftTopk}(\mathbf{a})^{(k)} = \mathrm{SoftTopk}(\mathbf{a})^{(k-1)} + \mathrm{SE}(\mathbf{a}),$$

$$\mathrm{SE}(\mathbf{a})_i = \begin{cases} 0, a_i \in \mathrm{Topk}(\mathbf{a}) \\ t \cdot \mathrm{Softmax}(\mathbf{a})_i, \text{otherwise}, \end{cases}$$



- MaxScore has greatly improved the load balancing.



❖ Experiment

Main Results

Model	ARC challenge	ARC easy	BoolQ	Hella-Swag	LAM-BADA	PIQA	RACE	SciQ	Record	OBQA	Avg.
Dense	18.69	40.19	57.06	28.91	16.28	63.71	25.65	64.2	56.05	15.0	38.57
GShard	18.86	**44.49**	61.90	31.74	21.54	66.38	**28.52**	69.4	62.08	16.2	42.11
GShard-I	19.80	44.36	59.94	32.54	21.52	67.03	28.23	68.7	62.84	16.0	42.10
SBASE	18.34	43.73	57.61	30.96	19.70	65.18	27.37	68.3	60.06	16.2	40.75
ExpertChoice	19.37	42.00	61.74	32.10	21.19	66.16	27.18	68.4	62.26	17.6	41.80
DropLess	19.28	44.07	61.16	32.03	21.35	67.14	27.08	67.9	61.55	16.0	41.76
DeepSeek	19.88	44.28	60.55	32.23	21.93	66.97	27.94	70.9	62.57	17.6	42.49
MaxScore-I	**20.90**	43.22	61.71	32.51	21.66	**67.41**	28.42	69.9	63.61	**18.4**	42.77
MaxScore	20.73	**44.49**	**62.23**	**32.85**	**23.27**	**67.41**	**28.52**	**72.5**	**64.00**	**18.4**	**43.44**

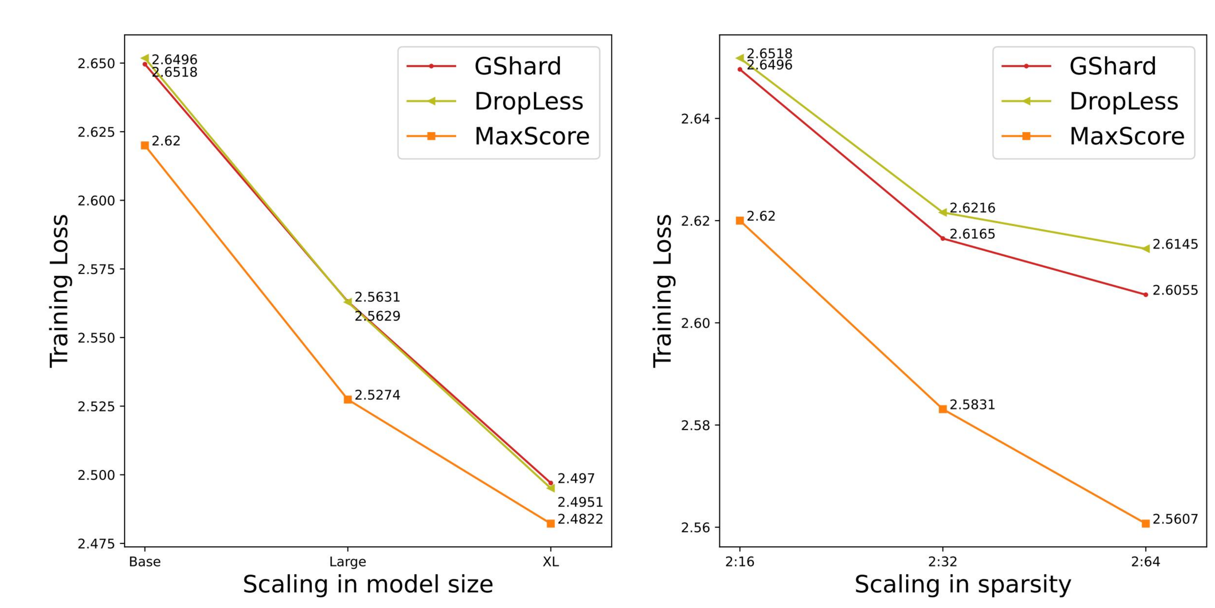
Abation Results

Model	ARC challenge	ARC easy	BoolQ	Hella-Swag	LAM-BADA	PIQA	RACE	SciQ	Record	OBQA	Avg.
GShard	18.86	**44.49**	61.90	31.74	21.54	66.38	**28.52**	69.4	62.08	16.2	42.11
GShard-I	19.80	44.36	59.94	32.54	21.52	67.03	28.23	68.7	62.84	16.0	42.10
GShard-M	20.14	43.74	59.38	32.27	22.30	66.63	27.61	68.7	62.59	18.2	42.16
GShard-S	20.52	44.30	59.13	32.34	22.54	66.74	28.19	69.4	63.94	**18.4**	42.55
GShard-SI (MaxScore-I)	**20.90**	43.22	61.71	32.51	21.66	**67.41**	28.42	69.9	63.61	**18.4**	42.77
GShard-SM (MaxScore)	20.73	**44.49**	**62.23**	**32.85**	**23.27**	**67.41**	**28.52**	**72.5**	**64.00**	**18.4**	**43.44**

Scalability Results

Model	Base	Large	XL
Activated Params	$162M$	$317M$	$600M$
Total Params	$757M$	$1.6B$	$3.2B$
FLOPs	$302G$	$603G$	$1.2T$

Sparsity	2:16	2:32	2:64
Activated Params	$162M$	$162M$	$162M$
Total Params	$757M$	$1475M$	$2867M$
FLOPs	$302G$	$302G$	$302G$



Size	Model	ARC challenge	ARC easy	BoolQ	Hella-Swag	LAM-BADA	PIQA	RACE	SciQ	Record	OBQA	Avg.
Base	**GShard**	18.86	**44.49**	61.90	31.74	21.54	66.38	**28.52**	69.4	62.08	16.2	42.11
	DropLess	19.28	44.07	61.16	32.03	21.35	67.14	27.08	67.9	61.55	16.0	41.76
	MaxScore	**20.73**	**44.49**	**62.23**	**32.85**	**23.27**	**67.41**	**28.52**	**72.5**	**64.00**	**18.4**	**43.44**
Large	**GShard**	19.88	45.58	62.16	33.34	23.69	67.74	29.28	70.0	64.99	19.2	43.59
	DropLess	20.05	45.24	61.19	33.97	23.60	67.63	27.66	69.7	63.95	17.2	43.02
	MaxScore	**20.90**	**45.92**	**62.39**	**34.00**	**24.96**	**68.28**	**29.67**	**74.3**	**66.12**	**19.8**	**44.63**
XL	**GShard**	20.05	46.68	63.09	35.14	25.05	69.31	29.19	72.7	67.50	20.0	44.87
	DropLess	20.14	46.60	61.69	35.11	24.34	68.34	29.19	72.4	67.55	20.2	44.56
	MaxScore	**21.22**	**47.60**	**63.60**	**35.57**	**25.93**	**69.95**	**29.90**	**75.2**	**67.90**	**21.6**	**45.85**

(a) Scaling in model size.

Sparsity	Model	ARC challenge	ARC easy	BoolQ	Hella-Swag	LAM-BADA	PIQA	RACE	SciQ	Record	OBQA	Avg.
2:16	**GShard**	18.86	**44.49**	61.90	31.74	21.54	66.38	**28.52**	69.4	62.08	16.2	42.11
	DropLess	19.28	44.07	61.16	32.03	21.35	67.14	27.08	67.9	61.55	16.0	41.76
	MaxScore	**20.73**	**44.49**	**62.23**	**32.85**	**23.27**	**67.41**	**28.52**	**72.5**	**64.00**	**18.4**	**43.44**
2:32	**GShard**	19.62	44.57	62.28	32.63	21.99	67.19	**29.04**	69.6	62.77	18.2	42.79
	DropLess	19.62	44.51	62.23	32.36	21.79	67.10	27.46	68.3	62.20	16.8	42.24
	MaxScore	**20.90**	**44.60**	**63.73**	**33.24**	**23.76**	**67.63**	**29.04**	**73.5**	**64.41**	**18.8**	**43.96**
2:64	**GShard**	19.80	44.86	62.26	33.05	22.20	67.30	28.46	69.4	63.17	17.6	42.81
	DropLess	19.60	44.69	62.40	32.79	21.65	67.27	27.56	69.5	63.05	17.6	42.61
	MaxScore	**21.11**	**46.17**	**64.24**	**33.38**	**23.41**	**67.95**	**28.90**	**73.3**	**64.60**	**19.0**	**44.21**

(a) Scaling in sparsity.

❖ Take away

- 1. Point out the reasons why the iterative method and the Sinkhorn method do not improve the model quality in MoE routing.
- Introduce the **Soft-topk** operator into the MoE routing.
- Through the **Minimum-Cost Maximum-Flow** modeling, we designed a MoE routing mechanism that has almost no additional computational cost and improves the quality of the model.

Code:
https://github.com/dongbw18/MaxScore

Paper:

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